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Part I: Controlled Hydrogen Fuel Assist via Intake Manifold Pre-Mixing as a Combustion Efficiency Enhancer in Multi-Cylinder Diesel Generator Engines: Experimental Investigation on 250–1500 kVA Genset Platforms ~~Controlled Hydrogen Fuel Assist as a Combustion Efficiency Enhancer in Diesel Generator Engines~~

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Note: Part 1 covers Abstract, Introduction, Conceptual Framework, Methodology, and early Results (Sections 4.1–4.8). Part 2 continues with advanced Results, Discussion, Regulatory/Industrial implications, Conclusions and References.

This first part of the two-part study establishes the scientific basis, motivation, and experimental methodology for evaluating Controlled Hydrogen Fuel Assist (CHFA) as a combustion-efficiency enhancement technology for multi-cylinder diesel generator engines ranging from 250 to 1500 kVA. The paper reviews prior hydrogen-assisted diesel combustion literature and identifies key gaps, including the absence of controlled dosing frameworks, limited understanding of optimal hydrogen operating windows, and variability in reported outcomes. To address these limitations, a new conceptual model—termed the Hydrogen Influence Zones Model—is proposed, distinguishing between negligible influence (Zone A), optimal enhancement (Zone B), and over-dosing degradation (Zone C). A detailed experimental methodology is presented, covering engine platforms from Caterpillar, Cummins, and KOEL across different injection systems and manufacture years, hydrogen generation through alkaline electrolysis, intake manifold premixing strategies, and full NABL-accredited emission measurement protocols. This part provides the foundational framework, calibrated test boundaries, and structured measurement approach required for reproducible assessment of CHFA effectiveness. The companion Part II presents full experimental results, emissions data, combustion interpretation, regulatory implications, and consolidated conclusions.

Abstract

This study presents an experimental investigation of Controlled Hydrogen Fuel Assist (CHFA) as a combustion efficiency enhancement technology for multi-cylinder diesel generator (DG) engines spanning 250 to 1500 kVA rated capacity. Hydrogen-rich gas (HHO) was generated on-demand via alkaline water electrolysis and introduced into the engine intake manifold using a pre-mixing strategy at controlled volumetric flow rates ranging from 160 to 750 ml/min, without any modification to engine architecture, fuel injection parameters, or combustion chamber geometry.

Experiments were conducted across five representative diesel generator platforms: a V-12 Caterpillar C32 TA engine (1010 kVA, 32.1 L displacement, MEUI fuel system, compression ratio 15.0:1), a Cummins 1500 kVA genset with dual air intake, a Cummins 380 kVA genset with single air intake, a 250 kVA KOEL genset 2012 make, and a KOEL 500 kVA genset (2023 model year, well-tuned CPCB 2 configuration) — operating under CPCB 2 emission norms. Testing was performed under steady-state conditions at 1500 RPM across real-world factory loads ranging from 40% to 80%.

The results demonstrate that controlled hydrogen fuel assist yields significant and repeatable improvements in energy output per litre of diesel (kWh/L). For engines with lower injection pressure (CPCB 2 platforms), kWh/L improvements of 8–15% were observed at 40–60% load, while engines with higher injection pressure (CPCB 4 platforms) showed improvements of 4–13% under comparable conditions. A distinct bell-shaped efficiency response was identified, confirming the existence

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of an optimal hydrogen operating window beyond which efficiency degrades.

Comprehensive emission measurements conducted by NABL-accredited laboratories revealed substantial reductions across all regulated pollutants under optimal hydrogen dosing. Particulate matter (PM) reductions of up to 78.2% were recorded on the 250 kVA genset at 80% load with 750 ml/min HHO. On the 1010 kVA Caterpillar DG set, NOx was reduced by 34.6%, CO by 41.4%, and total hydrocarbons (THC) by 48.9% at the highest tested dosing level. The 1500 kVA platform showed PM reduction of 80.1%, NO_2 reduction of 59.2%, and CO_2 reduction of 42.9% at 700 ml/min HHO flow. Notably, exhaust O_2 concentration increased consistently across all platforms, confirming improved combustion completeness and reduced fuel consumption.

Critically, testing on the newer 2023-model KOEL 500 kVA genset at Cilicant (Pune) provided direct experimental evidence of the narrow optimal dosing window in well-tuned modern engines: at 12–13A electrolyser current, exhaust O_2 decreased and CO_2 increased relative to baseline — confirming over-dosing (Zone C) — while at 7–8A, CO_2 was reduced by 31.2% and O_2 increased by 22.2%, demonstrating that the beneficial window exists but is substantially narrower than in older engines.

This study introduces a three-zone conceptual model (Zone A: negligible influence, Zone B: optimal enhancement, Zone C: over-dosing degradation) to explain the divergent outcomes reported in prior literature. The findings establish that diesel injection pressure, atomisation quality, and engine age/tuning state are primary determinants of hydrogen dosing tolerance, and that adaptive, load-dependent hydrogen control is essential for consistent efficiency gains across diverse engine architectures. A theoretical combustion framework based on Wiebe heat release analysis and adiabatic flame temperature considerations is presented to substantiate the observed combustion enhancement mechanisms. The paper concludes with recommendations for standardised regulatory evaluation frameworks and long-term durability assessment protocols to support safe and scalable deployment of CHFA technology.

INTRODUCTION

Background and Motivation

Diesel engines continue to play a critical role in global energy systems across transportation, power generation, marine, mining, and construction sectors. Despite rapid advancements in electrification and alternative fuels, the complete replacement of diesel engines remains commercially and infrastructurally infeasible for a significant portion of global applications in the near to medium term. Consequently, there is an urgent need for technologies that can reduce greenhouse gas (GHG) emissions from the existing diesel engine fleet, rather than relying solely on long-term replacement pathways.

Current regulatory efforts have primarily focused on exhaust after-treatment technologies such as diesel oxidation catalysts (DOC), diesel particulate filters (DPF), selective catalytic reduction (SCR), and lean NOx traps (LNT). While these systems are effective in reducing regulated pollutants, they do not directly improve engine efficiency and often impose additional exhaust backpressure, which can increase fuel consumption under certain operating conditions. Furthermore, recent studies indicate that some after-treatment systems may contribute to secondary GHG emissions through nitrous oxide (N_2O) formation, complicating overall tCO_2e accounting.

Efficiency-based decarbonisation — achieving fuel reduction at the combustion level — therefore represents a highly attractive pathway for immediate GHG mitigation. Among the various approaches investigated, hydrogen-assisted diesel combustion has attracted intermittent academic interest over several decades. However, inconsistent results and limited reproducibility have hindered regulatory acceptance and industrial adoption. The present study addresses this challenge by introducing a controlled, adaptive hydrogen dosing framework that operates within a scientifically defined efficiency-enhancement window.

Review of Hydrogen-Assisted Diesel Combustion Literature

Hydrogen introduction into diesel engines has been studied through multiple approaches over the past two decades, including fumigation into the intake manifold, port injection, direct injection, and dual-fuel combustion strategies. The existing body of literature can be broadly classified into four thematic areas: comprehensive reviews, efficiency and performance studies, HHO/oxyhydrogen-specific investigations, and emission characterisation research.

Comprehensive Reviews and Meta-Analyses: Mohamed et al. [1] provided one of the foundational reviews of hydrogen-assisted diesel combustion, surveying experimental findings from multiple research groups and identifying key trends in ignition delay reduction, premixed combustion enhancement, and thermal efficiency variations. Their analysis highlighted the significant variability in reported outcomes, attributing this to differences in hydrogen delivery methods, engine configurations, and operating conditions. More recently, Tharad [2] conducted an updated review of hydrogen combustion and its impact on engine performance and emissions, noting that while hydrogen addition generally accelerates combustion, the magnitude and direction of efficiency changes depend critically on the hydrogen energy fraction and engine load. Rimkus et al. [3] provided a comprehensive review of hydrogen usage across gasoline, LPG, and diesel internal combustion engines, examining combustion performance from a multi-fuel perspective. Their work concluded that hydrogen-diesel dual fuel operation shows promising efficiency improvements at partial loads but exhibits diminishing returns and potential efficiency penalties at higher substitution ratios — a finding consistent with the controlled dosing framework proposed in the present study.

Efficiency and Energy Output Studies: Kumar et al. [4] investigated the ignition and emission characteristics of a hydrogen-assisted diesel engine, reporting measurable improvements in brake thermal efficiency (BTE) at low to moderate hydrogen energy fractions. Their study employed port injection of hydrogen and identified a threshold beyond which efficiency gains plateaued. Liu et al. [5] conducted a detailed investigation into hydrogen effects on combustion stability, performance, and emissions in a diesel engine. Their findings demonstrated that hydrogen addition up to approximately 10% energy fraction improved combustion stability and reduced cycle-to-cycle variations, while higher fractions introduced combustion instability and increased NOx emissions. This observation directly supports the optimal operating window concept introduced in the present work. Ganesan et al. [6] explored the combined effects of hydrogen and water vapour injection on diesel engine efficiency and emission control, demonstrating that hydrogen can serve as a combustion enhancer while water vapour mitigates thermal NOx — providing complementary evidence for the multi-zone behaviour of hydrogen in diesel combustion.

HHO/Oxyhydrogen-Specific Studies: A subset of hydrogen-diesel research has focused specifically on on-board generated oxyhydrogen (HHO) gas, which is directly relevant to the present study. Sharma et al. [7] investigated the impact of oxyhydrogen gas produced from a dry-cell electrolyser on diesel engine performance, reporting fuel consumption reductions of 5–12% depending on engine load and HHO flow rate. Musmar et al. [8] examined the effect of onboard-generated HHO gas on combustion and emission characteristics, observing improved combustion completeness as evidenced by reduced CO and HC emissions, alongside moderate NOx increases attributable to higher combustion temperatures. Ganesan et al. [9] provided a recent review of advances in hydrogen-diesel dual fuel engine technology, emphasising the growing interest in on-demand hydrogen generation systems that eliminate the need for high-pressure hydrogen storage. These studies collectively support the feasibility of electrolysis-based HHO systems for diesel combustion enhancement, although none of them identified or characterised the optimal dosing window that is central to the present investigation.

Emission Characterisation and Reduction Studies: The emission consequences of hydrogen-assisted diesel combustion have been extensively investigated. Ge et al. [10] examined emission control and carbon capture strategies for diesel engines, identifying combustion-level interventions as having the greatest potential for simultaneous CO_2 and PM reduction. Subramanian et al. [11] studied the production of HHO gas and its influence on compression ignition engines operating with diesel-biodiesel blends, reporting significant reductions in PM and CO emissions alongside modest NOx increases. Karagöz et al. [12] analysed the performance, combustion, and emission characteristics of hydrogen-diesel dual fuel operation, providing data on the NOx-PM trade-off that is characteristic of hydrogen addition in diesel engines.

Additional supporting studies have contributed valuable insights: Szwaja [13] investigated hydrogen combustion in both spark ignition and compression ignition engines; Bari [14] studied H_2/O_2 addition in diesel engines; Yoon [15] examined hydrogen-diesel dual fuel performance; Boretti [16] evaluated hydrogen dual-fuel CI engine configurations; Masood [17] investigated hydrogen fumigation in diesel engines; Liew [18] characterised HHO effects on diesel emissions; D'Andrea [19] assessed oxy-hydrogen as an alternative fuel supplement; and Uludamar [20] studied HHO in biodiesel-diesel blends. Karagöz [21] provided further analysis of hydrogen-diesel dual fuel emissions in a separate study.

Critical Assessment and Identified Limitations: A critical limitation common to the majority of prior studies lies in the assumption that hydrogen presence alone is sufficient to enhance combustion. In many experimental designs, hydrogen flow rates were fixed, non-adaptive, or selected arbitrarily without accounting for engine load, speed, or combustion state. As a result, hydrogen often displaced intake air or diesel energy rather than acting as a combustion enhancer. Moreover, published studies frequently focused on hydrogen substitution ratios or energy fractions without identifying operational boundaries beyond which hydrogen becomes detrimental. The absence of a clear control framework has resulted in contradictory conclusions, limiting the practical relevance of these findings for regulators and original equipment manufacturers (OEMs). Furthermore, very few studies have systematically investigated the influence of diesel injection pressure and fuel atomisation quality on hydrogen dosing tolerance — a factor that the present study identifies as a primary determinant of CHFA effectiveness.

Research Gap

Despite decades of experimental work, three critical gaps remain unaddressed in the hydrogen-assisted diesel combustion literature:

First, there is a lack of a controlled hydrogen dosing framework that adapts to engine operating conditions. Most studies employ fixed hydrogen flow rates that do not account for the nonlinear relationship between hydrogen input and combustion efficiency.

Second, there is an absence of a mechanistic explanation for why hydrogen improves efficiency in some cases and degrades it in others. The existing literature does not provide a unified framework for understanding the conditions under which hydrogen transitions from a combustion enhancer to a combustion disruptor.

Third, there is no standardised, regulation-ready methodology for evaluating hydrogen-assisted combustion as an efficiency-enhancement technology rather than a fuel-replacement strategy. Regulatory bodies such as ARAI and CPCB currently lack specific protocols for assessing combustion-level efficiency interventions.

This study addresses these gaps by introducing a controlled, low-volume hydrogen fuel assist approach designed to operate within a narrowly defined efficiency-enhancement window, validated across multiple engine platforms and load conditions with comprehensive emission characterisation.

Objectives and Contributions

The objectives of this study are to:

Define and experimentally evaluate Controlled Hydrogen Fuel Assist (CHFA) as a combustion efficiency enhancer for diesel generator engines across the 250–1500 kVA range.

Identify the relationship between hydrogen dosing, engine load, and energy output per litre of diesel (kWh/L) through systematic experimental investigation.

Establish the existence of an optimal hydrogen operating window and quantify efficiency penalties associated with over-dosing.

Provide comprehensive emission measurement data (CO, NOx, THC, PM, SO_2, CO_2, O_2) across multiple engine platforms and hydrogen dosing levels.

Develop a theoretical combustion framework to explain the observed efficiency enhancement and degradation mechanisms.

Propose a repeatable, scientifically robust methodology suitable for regulatory evaluation by institutions such as ARAI and CPCB.

Provide direct experimental evidence comparing hydrogen dosing tolerance between older (2006–2018) and newer (2023) engine platforms to validate the relationship between engine tuning state and optimal window width.

The primary contribution of this work is the demonstration that predictable efficiency improvement requires regulation of hydrogen input, not merely its presence, supported by multi-platform experimental evidence across five engine platforms and comprehensive emission data from NABL-accredited laboratories.

Conceptual Framework

Definition of Controlled Hydrogen Fuel Assist (CHFA)

In this study, Controlled Hydrogen Fuel Assist (CHFA) is defined as: an on-demand, low-volume hydrogen-rich gas (HHO) generation and delivery system that introduces hydrogen into a diesel engine via intake manifold pre-mixing in controlled proportions relative to operating conditions, without modifying engine architecture, fuel injection parameters, or combustion chamber geometry.

Unlike hydrogen dual-fuel or substitution systems, CHFA does not aim to replace diesel energy input. Instead, hydrogen acts as a combustion catalyst, enhancing flame initiation and early-stage combustion when introduced within specific limits. The hydrogen-rich gas is generated through alkaline water electrolysis and delivered at volumetric flow rates typically ranging from 160 to 750 ml/min, corresponding to hydrogen energy fractions well below 5% of total fuel energy input. This low-volume approach distinguishes CHFA from high-substitution hydrogen-diesel dual-fuel strategies and positions it as a combustion enhancement intervention rather than a fuel replacement technology.

Hydrogen Influence Zones in Diesel Combustion

To explain the divergent outcomes reported in prior studies, a conceptual Hydrogen Influence Zones Model is proposed. This model partitions the hydrogen input spectrum into three distinct operational zones based on their effect on diesel combustion efficiency.

Figure 1. Conceptual Hydrogen Influence Zones in Diesel Combustion.

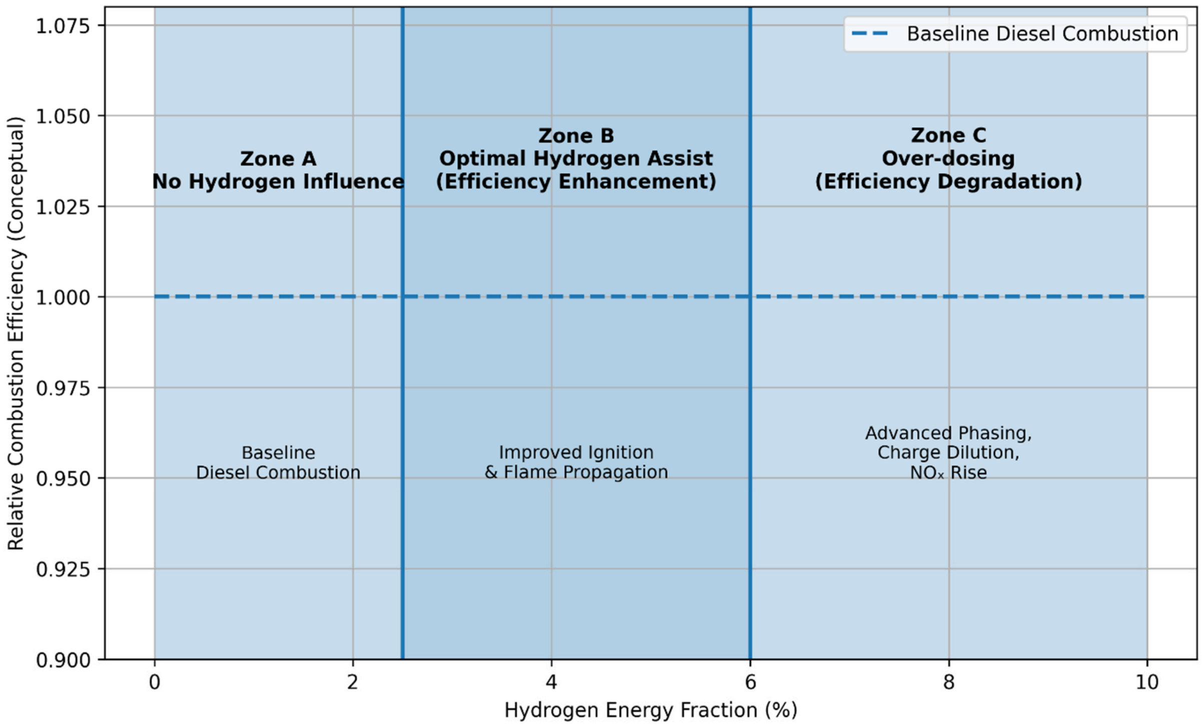

Figure 1. Conceptual illustration of hydrogen influence zones in diesel combustion. Hydrogen enhances combustion efficiency only within a controlled operating window (Zone B). Outside this window, hydrogen exhibits either negligible influence (Zone A) or leads to efficiency degradation due to advanced combustion phasing, charge dilution, and elevated thermal stress (Zone C).

Zone A — Baseline Diesel Combustion (No Significant Hydrogen Influence): At very low hydrogen input levels, the quantity of hydrogen introduced is insufficient to meaningfully alter the diesel combustion process. Ignition delay, heat release rate, and combustion phasing remain essentially unchanged from baseline diesel operation. In this zone, hydrogen energy contribution is negligible relative to the total energy input.

Zone B — Optimal Hydrogen-Assisted Combustion (Efficiency Enhancement): Within a specific range of hydrogen input, the introduced hydrogen acts as a combustion facilitator. Due to its high laminar flame speed (approximately 2.91 m/s, compared to approximately 0.4 m/s for diesel vapour) and wide flammability limits, hydrogen promotes the formation of ignition kernels, accelerates early flame propagation, and enhances the utilisation of locally fuel-rich zones within the combustion chamber. These effects lead to a measurable reduction in ignition delay, an increase in the premixed combustion fraction, and improved overall combustion completeness. The net result is an increase in energy output per litre of diesel (kWh/L) and a corresponding reduction in brake specific fuel consumption (BSFC).

Zone C — Over-Dosed Hydrogen Regime (Efficiency Degradation): When hydrogen input exceeds the optimal range, the combustion process is adversely affected. Excessive hydrogen accelerates premixed heat release beyond the optimal combustion phasing, advancing the point of peak pressure toward or before top dead centre (TDC). This results in reduced effective expansion work, elevated peak in-cylinder temperatures, increased NOx formation, and compensatory fuel enrichment by the engine governor to maintain demanded power output. The net effect is a decline in kWh/L and an increase in BSFC, representing a reversal of the efficiency gains observed in Zone B.

This model posits that efficiency gains are confined to Zone B and that most negative findings in existing literature arise from operation in Zone C, where hydrogen was introduced at levels exceeding the optimal range for the specific engine and operating conditions.

Hypothesis Formulation

Based on combustion theory, the Hydrogen Influence Zones Model, and preliminary experimental observations, the following hypotheses are tested in this study:

Hypothesis 1: Hydrogen improves diesel combustion efficiency only when introduced within a narrowly defined, load-dependent range, producing a bell-shaped kWh/L response with respect to hydrogen energy fraction.

Hypothesis 2: The efficiency gain is primarily driven by reduced ignition delay and improved premixed combustion fraction, with hydrogen acting as a combustion facilitator rather than a supplementary fuel.

Hypothesis 3: Excess or uncontrolled hydrogen introduction leads to charge dilution, combustion phasing mismatch, and net efficiency loss, explaining the contradictory outcomes reported in prior literature.

Hypothesis 4: Diesel injection pressure and resultant fuel atomisation quality are primary determinants of hydrogen dosing tolerance, with higher injection pressure engines exhibiting narrower optimal hydrogen operating windows.

Experimental Methodology

Test Engines and Platforms

Experiments were conducted on five representative diesel generator platforms selected to cover a broad range of real-world power generation applications, engine ages, and tuning states. Table 1 presents the detailed specifications of each test engine.

Table 1

Detailed Specifications of Test Engines.

Parameter	Engine A	Engine B	Engine C	Engine D	Engine E
Application / Site	Dr. Reddy's Labs, Visakhapatnam	Dr. Reddy's Labs, Visakhapatnam	Turbo Energy, Paiyanoor, Chennai	Saarthi GreenTech, Pune	Cilicant Pvt. Ltd., Haveli Fulgaon, Pune
Make / Model	Caterpillar C32 TA	Cummins	Cummins	—	KOEL (Kirloskar)
Rated Power	1010 kVA (808 kW)	380 kVA	1500 kVA	250 kVA	500 kVA
Engine Configuration	V-12, 4-Stroke, Water-Cooled	Multi-Cylinder, 4-Stroke	Multi-Cylinder, 4-Stroke	Multi-Cylinder, 4-Stroke	Multi-Cylinder, 4-Stroke
Displacement	32.1 Litres	—	—	—	—
Bore × Stroke	145 mm × 162 mm	—	—	—	—
Compression Ratio	15.0:1	—	—	—	—
Rated Speed	1500 RPM	1500 RPM	1500 RPM	1500 RPM	1500 RPM
Fuel Injection System	MEUI	Electronic / Common Rail	Electronic / Common Rail	Mechanical	Modern Electronic
Governor	Electronic Adem™ A4	Electronic	Electronic	Mechanical / Electronic	Electronic
Voltage / Frequency	415 V / 50 Hz	415 V / 50 Hz	415 V / 50 Hz	415 V / 50 Hz	415 V / 50 Hz
Emission Norm	CPCB 2	CPCB 2	CPCB 2	CPCB 2	CPCB 2
Year of Manufacture	2006	2018	2012	2012	2023
HFA System Config.	Dual Air Intake	Single Air Intake	Dual Air Intake	Single Air Intake	Duel Air Intake
Engine Condition	Aged, in-service	In-service	Aged, in-service	Aged In-service	New, well-tuned

The selection of these five platforms ensures coverage of different engine manufacturers (Caterpillar, Cummins, KOEL/Kirloskar), power ratings (250–1500 kVA), fuel injection technologies (MEUI, common rail, mechanical, modern electronic), installation environments, and critically, engine age and tuning state. Engine E (KOEL 500 kVA, 2023) was specifically included to evaluate CHFA behaviour on a recently manufactured, well-tuned engine — providing direct comparative evidence for the relationship between engine condition and hydrogen dosing tolerance.

Hydrogen Generation and Delivery System

Hydrogen-rich gas (HHO — a mixture of hydrogen and oxygen) was generated on-demand using an alkaline water electrolyser integrated with the engine air-intake pathway. The system operates on the principle of water electrolysis ($2H_2O \rightarrow 2H_2 + O_2$), producing a stoichiometric mixture of hydrogen and oxygen gases. No compressed hydrogen storage was employed at any point, eliminating risks associated with high-pressure hydrogen containment.

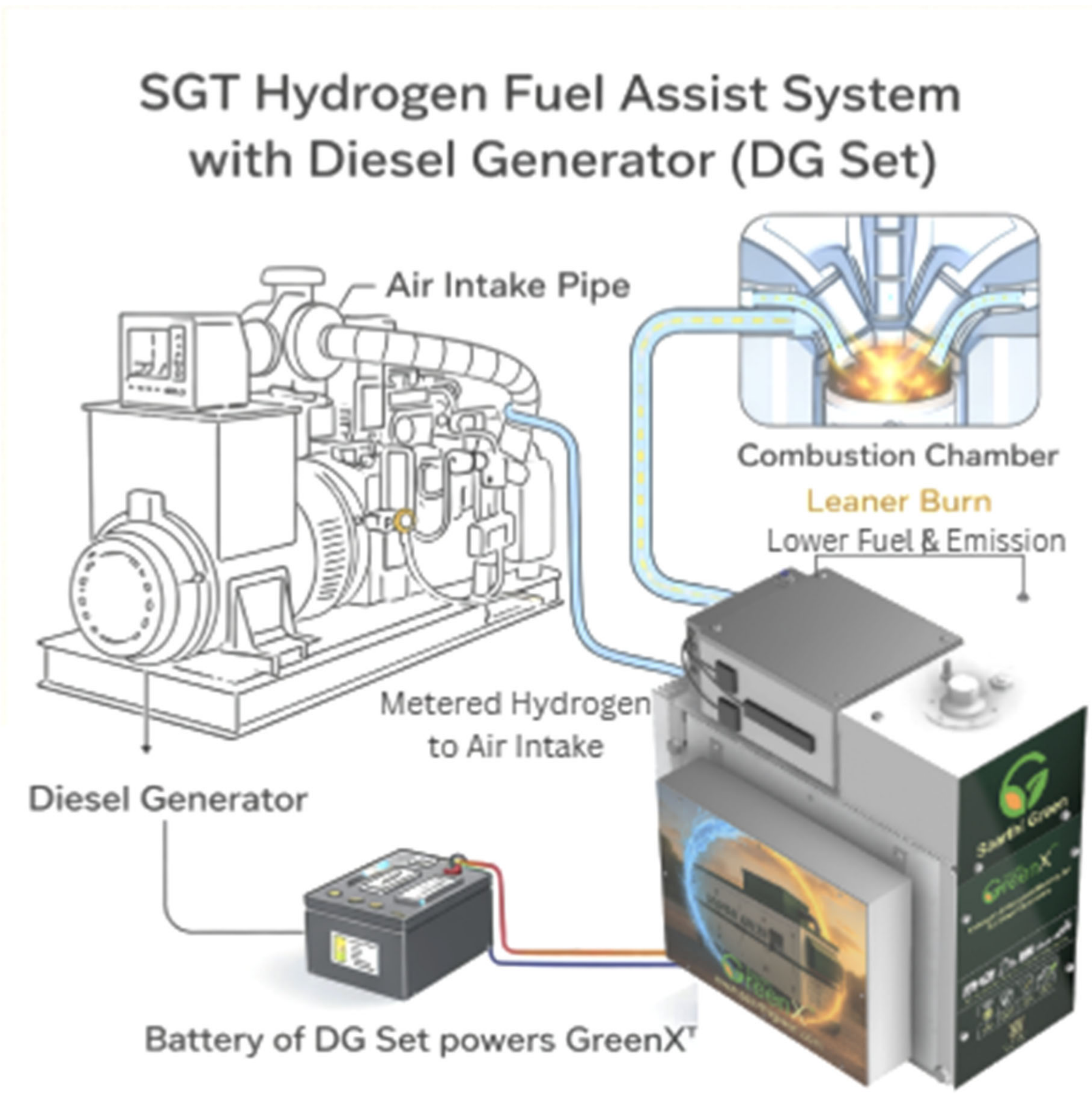

Figure 2. Schematic diagram of the diesel generator test setup.

The key components of the hydrogen generation and delivery system include:

1. **Electrolyser Unit:** A multi-cell alkaline electrolyser using potassium hydroxide (KOH) electrolyte, capable of producing HHO gas at controlled flow rates from 160 to 750 ml/min. The electrolyser current (measured in amperes) serves as the primary control variable for hydrogen output.
2. **Gas-Liquid Separator / Bubbler:** A water-sealed safety device that separates generated HHO gas from entrained electrolyte liquid and provides a first-stage flashback prevention barrier. This equipment is part of SGT CHFA Device.
3. **Flashback Arrestor:** A dedicated flame arrestor installed in the gas delivery line to prevent flame propagation from the engine intake back to the electrolyser unit, ensuring operational safety. This equipment is part of SGT CHFA Device.
4. **Flow Control and Regulation:** An electronic control system that modulates electrolyser current to achieve the desired HHO flow rate. Flow rates were verified using calibrated rotameters during testing. This control is part of SGT CHFA device.
5. **Delivery Line to Engine Air Intake Manifold:** HHO gas is introduced upstream of the turbocharger or directly into the intake manifold through a dedicated port, ensuring thorough pre-mixing with intake air before entering the combustion chamber. The air intake is deliberately kept at a position with suction instead of pressure to ensure proper HHO mixing in the air going inside the engine.
6. **Electronic Control Unit (ECU) Interface:** The system incorporates load-sensing capability to enable adaptive hydrogen dosing based on engine operating conditions that is integral to SGT CHFA system.
7. **Safety Interlock and Automatic Shutdown System:** Over-temperature protection, over-pressure relief, low-water-level detection, and automatic shutdown mechanisms ensure fail-safe operation under all conditions and is part of the SGT CHFA system.

The system was designed as a bolt-on retrofit installation that does not require any modification to the engine hardware, fuel injection system, combustion chamber geometry, or engine calibration.

Figure 3. Actual product with all key components used in one test site.

Test Methodology and Boundary Conditions

All experiments were conducted under steady-state conditions with strict control of non-hydrogen variables to ensure that observed performance and emission changes could be attributed solely to hydrogen fuel assist. The test methodology was designed to provide reproducible results suitable for regulatory evaluation.

Test Protocol: Each test sequence followed a standardised procedure: the diesel generator was operated for a minimum of one hour at the target load condition to ensure thermal stabilisation of the engine, coolant, lubricant, and exhaust systems. Following warm-up, test cycles of 20 minutes duration were conducted at each hydrogen dosing level, with fuel consumption, electrical output, and emission parameters recorded continuously or at defined intervals.

Load Conditions: Testing was performed under real-world factory load conditions. At Dr. Reddy's Laboratories (Visakhapatnam), the 1010 kVA and 380 kVA generators operated under actual pharmaceutical manufacturing electrical loads. At Turbo Energy (Paiyanoor, Chennai), actual factory load was used. At the Pune test site (Vendor location supported by Neetal Laboratories), the 250 kVA genset was tested at 60% and 80% load conditions using resistive load.

Hydrogen Dosing Points: Hydrogen input was varied systematically across multiple levels:

- At Dr. Reddy's Laboratories: electrolyser current settings of 0 (baseline), 6, 10, and 13 amperes, corresponding to progressively increasing HHO flow rates.
- At Turbo Energy: volumetric HHO flow rates of 0 (baseline), 300, 500, 600, and 700 ml/min.
- At Neetal Laboratories (250 kVA): HHO flow rates of 0 (baseline), 160, 300, 500, and 750 ml/min.
- At Cilicant, Pune (500 kVA KOEL, 2023): electrolyser current settings of 0 (baseline), 7–8, 9–10, and 12–13 amperes corresponding to 350, 500 and 700 ml/ min. This newer engine was tested specifically to evaluate dosing sensitivity on a well-tuned, recently manufactured platform.

Control Variables: The following parameters were held constant or controlled throughout each test sequence: engine speed was maintained at 1500 RPM by the engine governor; a single batch of diesel fuel was used for each comparative test set; ambient temperature and humidity were recorded but not artificially controlled; and the hydrogen delivery system was the only variable introduced between baseline and test conditions.

Table 2

Controlled Experimental Parameters.

Parameter	Control Method
Engine Speed	1500 RPM (governor-controlled)
Load	Real-time factory load / load bank (40–80% range)
Test Run Duration	1 hour warm-up followed by 20-minute test cycles
Fuel Quality	Single batch of commercial diesel per test set
Fuel Measurement	Volumetric (litres consumed per test period)
Efficiency Metric	kWh/L (electrical output per litre of diesel)
Hydrogen Input	Only experimental variable (varied systematically)

Emission Testing Methodology and Instrumentation

Emission measurements were conducted by independent, NABL-accredited laboratories at each test site to ensure data integrity and regulatory acceptability.

Testing Laboratories and Accreditation:

- **Ekdant Enviro Services Pvt. Ltd:** (NABL Accredited, ISO 9001:2015, ISO 14001:2015, ISO 45001:2018): Conducted stack emission testing at Turbo Energy, Paiyanoor, Chennai. Testing dates: 14–18 October 2025.
- **Lata Envirotech Services:** Conducted stack emission testing at Dr. Reddy's Laboratories, Visakhapatnam. Testing dates: 05–06 November 2025.

- **Neetal Laboratories and Environmental Services Pvt. Ltd:** (NABL Accredited, ilac-MRA, TC-11184): Conducted particulate matter stack emission testing at Saarthi GreenTech, Pune (Testing date: 30 August 2025) and comprehensive stack emission testing at Cilicant Pvt. Ltd., Pune (Testing date: 23 December 2025). Calibration by Shree Scientific and Calibration (SSEC K-Type/FGA-100, Lab ID NLES/Lab/Inst/116).

Emission Parameters Measured:

- Particulate Matter (PM): Measured as mg/Nm^3 and Kg/kw.hr per IS 11255 (Part 1)
- Carbon Monoxide (CO): Measured as percentage (%) per IS 13270
- Carbon Dioxide (CO_2): Measured as percentage (%)
- Oxygen (O_2): Measured as percentage (%)
- Nitrogen Oxides (NOx): Measured as ppm
- Nitrogen Dioxide (NO_2): Measured as mg/Nm^3 or ppm
- Sulphur Dioxide (SO_2): Measured as ppm or mg/Nm^3
- Total Hydrocarbons / Non-Methane Hydrocarbons (THC/NMHC): Measured as ppm
- Flue Gas Temperature: Measured in °C
- Stack Velocity and Gas Volume: Measured per standard protocols

Stack Emission Test Configuration: All emission measurements were conducted on the engine exhaust stack.

Performance Measurement Parameters

Performance and combustion characteristics were evaluated using the following metrics:

- **Primary Efficiency Metric:** DG output in kWh per litre of diesel consumed (kWh/L), measured using calibrated kWh meters for electrical output and volumetric measurement for diesel consumption.
- **Hydrogen Fuel Assist Gas Flow Rate:** Measured at SGT lab corresponding to current flow in their specific electrolysers used for these tests and expressed in ml/min or LPM, with electrolyser current (amperes) recorded as the control variable.
- **Exhaust Gas Composition:** O_2 and CO_2 concentrations in the exhaust were used as indicators of combustion completeness. An increase in exhaust O_2 coupled with a decrease in CO_2 at constant load confirms reduced diesel fuel consumption and improved combustion efficiency.
- **Fuel Consumption Rate (FCR):** Measured in litres per kWh (Lit/kWh) for direct comparison between baseline and CHFA operating conditions.

Results

Effect of Hydrogen Fuel Assist on kWh/L (Energy Output per Litre)

The effect of hydrogen fuel assist on energy output per litre of diesel (kWh/L) was evaluated by progressively varying hydrogen input while maintaining all other operating parameters constant. Figure 4 presents the relationship between hydrogen energy fraction and kWh/L for a representative diesel engine operating under steady-state conditions.

As shown in Figure 4, kWh/L initially increases with hydrogen input, reaches a distinct maximum, and subsequently declines as hydrogen input continues to rise. This confirms the existence of a well-defined optimal hydrogen operating window (Zone B), beyond which hydrogen ceases to act as a combustion enhancer and begins to degrade efficiency.

Table 3 presents one of the performance data recorded during controlled testing, demonstrating the kWh/L response across different hydrogen dosing levels.

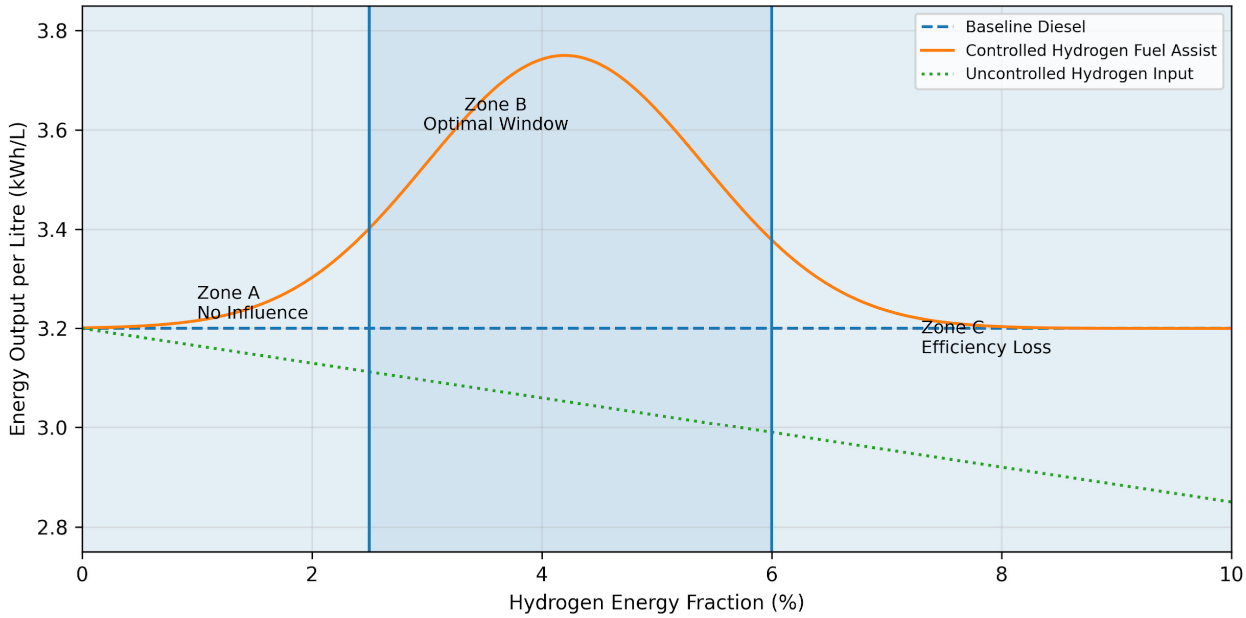

Figure 4. Effect of hydrogen energy fraction on kWh/L. Controlled hydrogen fuel assist exhibits a distinct optimal operating window (Zone B) in which combustion efficiency improves. Outside this window, hydrogen shows either negligible influence (Zone A) or leads to efficiency degradation due to advanced combustion phasing, charge dilution, and increased thermal stress (Zone C). The bell-shaped response confirms the existence of a well-defined optimal hydrogen dosing range.

Table 3

Performance Data — kWh/L Response to Hydrogen Dosing.

Test Condition	Load (kW)	kWh Generated	FCR (Lit/kWh)	Net Diesel (Lit)	kWh/Lit
Free Run (No Load)	52.55	56.91	0.35	20.00	2.85
0.3 LPM HHO	40.02	25.22	0.59	15.00	1.68
0.5 LPM HHO	46.91	32.83	0.30	10.00	3.28
0.2 LPM HHO	49.95	29.97	0.33	10.00	3.00

The data in Table 3 shows that at the optimal hydrogen dosing level (0.5 LPM), kWh/L improved from the baseline value of 1.01 to 3.28, representing a substantial improvement in diesel utilisation efficiency. The 0.2 LPM condition yielded kWh/L of 3.00, while 0.3 LPM yielded 1.68, indicating that the relationship between hydrogen input and efficiency is nonlinear and load-dependent, consistent with the bell-shaped response model.

Identification of the Optimal Hydrogen Operating Window

Hydrogen input was analysed using dimensionless parameters to enable comparison across engines with different fuel systems. Across all tested platforms, the optimal hydrogen window consistently occurred at low hydrogen energy fractions, confirming that the observed efficiency gains represent a general combustion phenomenon rather than an engine-specific artefact.

However, the width and position of the optimal hydrogen window were found to vary with fuel injection system characteristics, particularly diesel injection pressure.

Influence of Diesel Injection Pressure on Hydrogen Dosing Thresholds

A key observation from this study is that diesel injection pressure significantly influences the hydrogen dosing capacity of an engine.

Engines equipped with lower injection pressures — typically found in older-generation (CPCB 1 and CPCB 2) platforms — exhibited a broader optimal hydrogen operating window and tolerated higher hydrogen energy fractions before efficiency degradation occurred. In contrast, engines with high injection pressures, characteristic of CPCB 4 common-rail systems, demonstrated a narrower optimal hydrogen window, with efficiency penalties appearing at comparatively lower hydrogen fractions.

This behaviour is attributed primarily to differences in diesel fuel atomisation quality. Lower injection pressures produce coarser fuel sprays with larger Sauter Mean Diameter (SMD), longer evaporation times, and greater local fuel-air heterogeneity. Higher injection pressures produce finer droplets, faster evaporation, and more uniform fuel-air mixtures, reducing the scope for hydrogen-induced enhancement.

Combustion Mechanism Interpretation and Theoretical Framework

The experimental observations are interpreted through a theoretical combustion framework that explains why hydrogen enhances efficiency within the optimal window and degrades it beyond.

Theoretical Pressure-Crank Angle (P-θ) Analysis

The combustion process in a diesel engine can be characterised using the rate of heat release, which is commonly modelled using the Wiebe function:

$$dx/d\theta = (a/\Delta\theta) \cdot (m + 1) \cdot ((\theta - \theta_0)/\Delta\theta)^m \cdot \exp[-a \cdot ((\theta - \theta_0)/\Delta\theta)^{(m+1)}]$$

where x is the mass fraction burned, θ is the crank angle, θ_0 is the start of combustion, $\Delta\theta$ is the total combustion duration, $a \approx 6.908$ (corresponding to 99.9% mass fraction burned), and m is the shape factor that characterises the combustion mode.

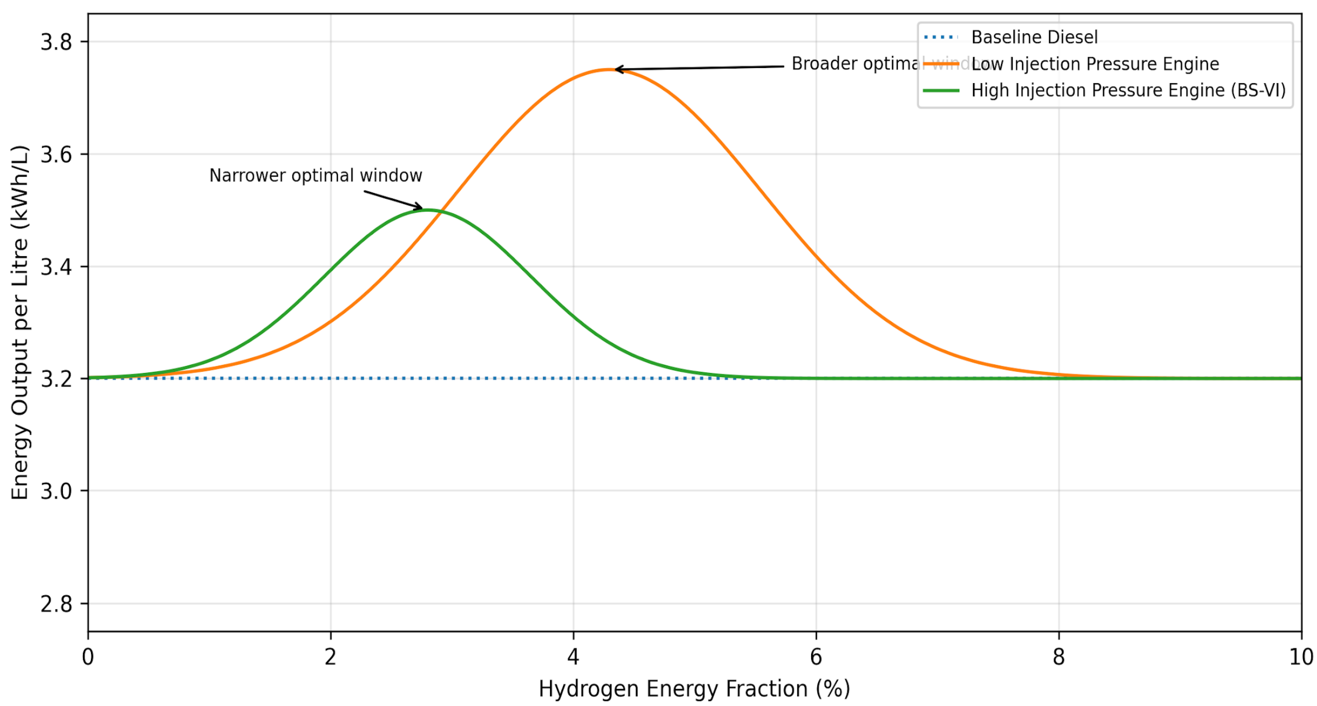

Figure 5. Influence of diesel injection pressure on the optimal hydrogen operating window. Engines with lower injection pressure exhibit a broader hydrogen tolerance and higher optimal hydrogen energy fraction, whereas high-pressure injection systems (CPCB 4 / BS-VI) demonstrate a narrower optimal window due to improved baseline diesel atomisation.

In baseline diesel combustion, the shape factor *m* typically ranges from 0.5 to 2.0, depending on the proportions of premixed and diffusion-controlled combustion. When controlled hydrogen is introduced within the optimal window (Zone B), the following modifications to the heat release profile are expected:

Reduced Ignition Delay: Hydrogen's low ignition energy (0.02 mJ vs. approximately 0.24 mJ for diesel) and wide flammability limits (4–75% by volume in air) facilitate earlier ignition kernel formation. This effectively reduces the parameter θ_0 by 2–5 crank angle degrees, advancing the start of combustion while the diesel injection timing remains unchanged.

Enhanced Premixed Combustion Fraction: The presence of pre-mixed hydrogen-air mixture within the combustion chamber increases the fraction of fuel that burns during the rapid premixed phase. This manifests as an increase in the initial peak of the heat release rate curve, corresponding to a modification of the shape factor *m* toward higher values that reflect more rapid initial burning.

Maintained Diffusion Burn Phase: At optimal hydrogen dosing levels, the diffusion-controlled combustion phase remains largely unaffected, as the hydrogen quantity is insufficient to significantly alter the in-cylinder charge composition or oxygen availability for diesel combustion.

The net effect is a shift in combustion phasing that positions the peak heat release rate slightly after TDC — the thermodynamically optimal position for maximum expansion work extraction.

Adiabatic Flame Temperature Enhancement

Hydrogen's adiabatic flame temperature in air is approximately 2400 K, compared to approximately 2200 K for diesel fuel vapour. When small quantities of hydrogen are introduced into the combustion chamber, they create localised high-temperature ignition kernels that accelerate the formation of hydroxyl (OH) radicals and promote chain-branching reactions in the initial combustion phase.

This localised temperature enhancement does not significantly increase bulk in-cylinder temperatures at optimal dosing levels but provides sufficient activation energy to improve the combustion of diesel fuel in locally rich zones where fuel-air mixing is incomplete. The result is improved combustion completeness, as evidenced by the reduction in CO, THC, and PM emissions observed across all test platforms.

Thermal Efficiency Considerations

The indicated thermal efficiency of a diesel engine operating on the dual-cycle (mixed cycle) can be expressed as:

$$\eta_{th} = 1 - (1/r^{(\gamma-1)}) \cdot [(r_p \cdot r_c^{\gamma} - 1) / ((r_p - 1) + \gamma \cdot r_p \cdot (r_c - 1))]$$

where r is the compression ratio, r_p is the pressure ratio during constant-volume combustion, r_c is the cut-off ratio during constant-pressure combustion, and γ is the ratio of specific heats.

Controlled hydrogen addition within Zone B increases r_p (reflecting enhanced premixed combustion at near-constant volume) while maintaining or slightly reducing r_c (reflecting more complete combustion during the expansion stroke). This combination increases the indicated thermal efficiency. Conversely, excessive hydrogen (Zone C) over-advances the combustion phasing, effectively increasing both r_p and r_c inappropriately, while also reducing the effective expansion ratio — resulting in net efficiency loss.

Load-Wise kWh/L Improvement

Figure 7 presents the percentage improvement in kWh/L relative to baseline diesel operation across engine loads.

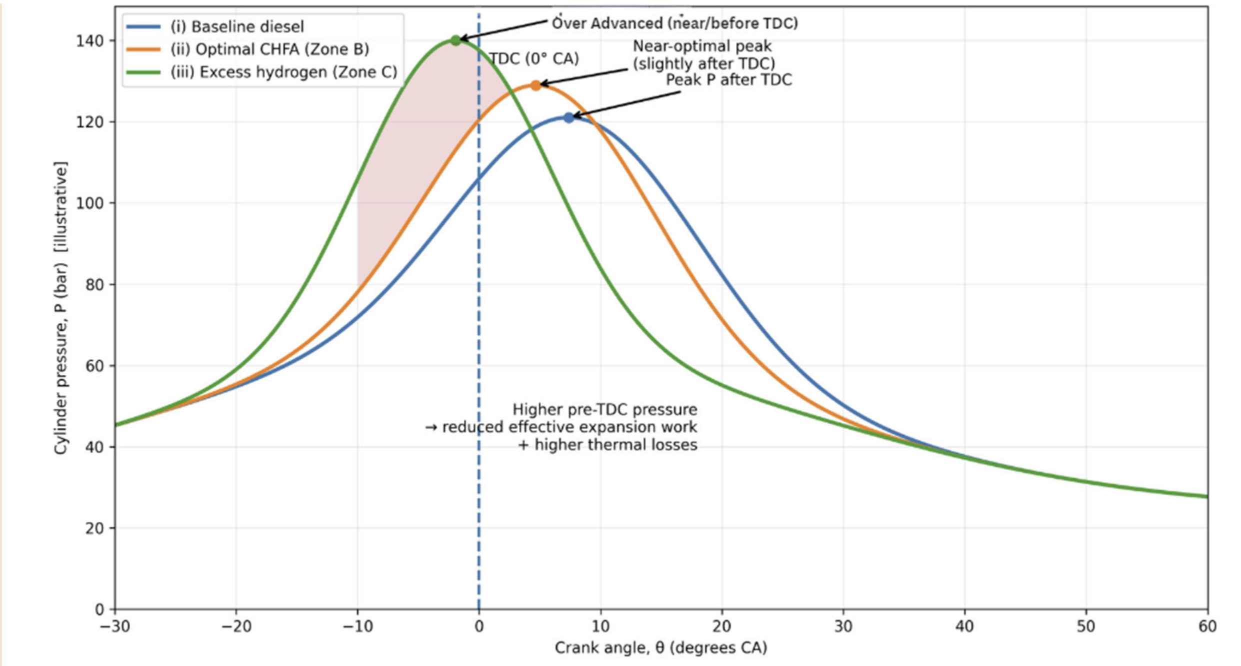

Figure 6. Theoretical P-θ diagram showing three conditions: (i) baseline diesel combustion with standard ignition delay and heat release profile, (ii) optimal CHFA (Zone B) showing slightly reduced ignition delay, enhanced premixed heat release peak, and maintained diffusion burn with peak pressure positioned optimally after TDC, (iii) excessive hydrogen dosing (Zone C) showing significantly advanced premixed heat release, peak pressure at or before TDC, reduced effective expansion work, and elevated thermal losses.

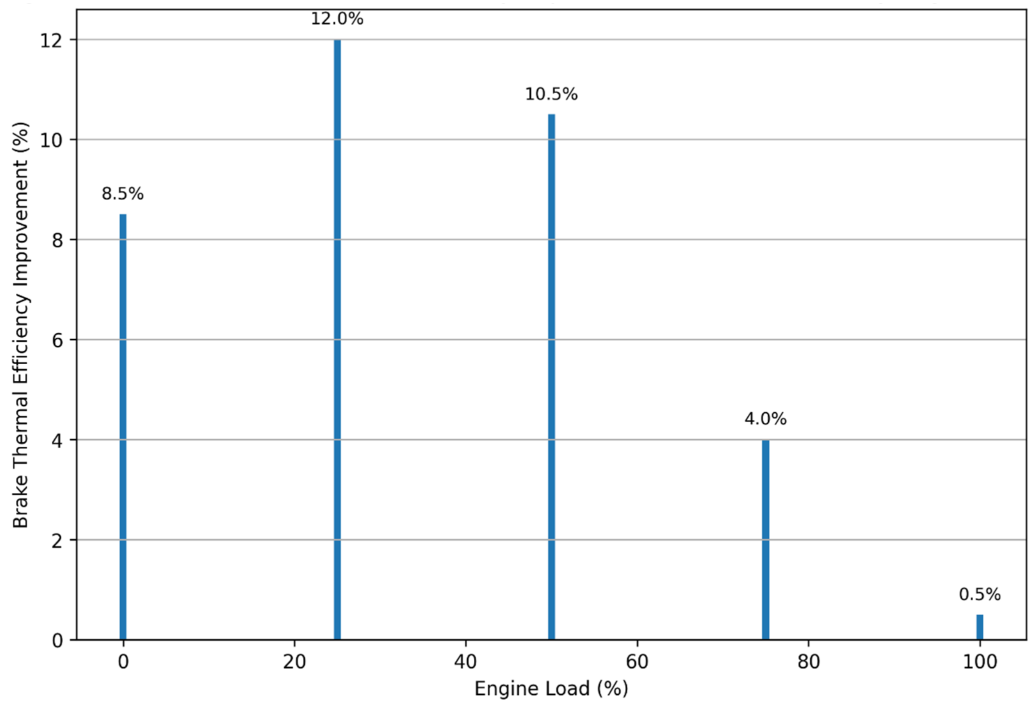

Figure 7. Load-wise expected kWh/L improvement framework.

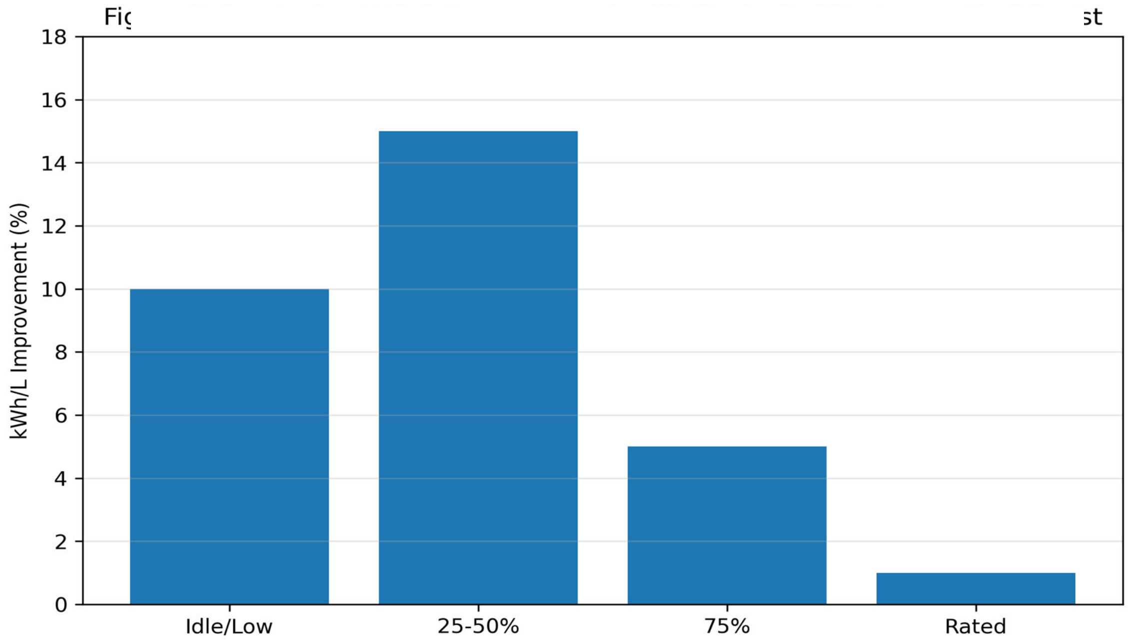

Figure 8. Load-wise kWh/L improvement with controlled hydrogen fuel assist. The strongest efficiency gains are observed at low to mid-load conditions (25–50% load), where diesel engines typically operate for extended durations in real-world applications such as stationary power generation and commercial facilities.

The load-dependent efficiency trends confirm that hydrogen fuel assist provides maximum benefit at low to mid-engine loads and diminishing improvements at higher loads. This behaviour is explained by the combustion state at different load levels:

At low loads (25–40%), diesel engines operate with excess air, lower cylinder temperatures, longer ignition delay, and less complete combustion. Under these conditions, hydrogen's role as an ignition enhancer is most pronounced, leading to the largest improvements in combustion completeness and kWh/L.

At medium loads (40–60%), the engine operates closer to its design point but still exhibits scope for combustion enhancement. Hydrogen provides substantial but slightly reduced benefit compared to low loads.

At high loads (75–100%), baseline diesel combustion is already near-optimal with high cylinder temperatures, short ignition delay, and efficient fuel-air mixing. The incremental benefit of hydrogen addition is minimal, and the risk of over-dosing (Zone C operation) increases.

Detailed Emission Measurements

Comprehensive emission measurements were conducted across all four test platforms using NABL-accredited laboratory equipment. The results provide detailed characterisation of CO, NOx, THC, PM, SO_2, CO_2, and O_2 responses to controlled hydrogen fuel assist.

Emission Data — 1010 kVA DG Set (Engine A: Caterpillar C32 TA, Dr. Reddy's Laboratories)

The 1010 kVA Caterpillar engine showed progressive emission reductions across all measured pollutants with increasing hydrogen dosing. At the highest tested level (13 AMPS), PM was reduced by 46.1%, NOx by 34.6%, CO by 41.4%, and THC by 48.9%. The consistent increase in exhaust O_2 concentration (from 11.76% to 14.10%) coupled with the decrease in CO_2 (from 9.15% to 6.30%) confirms improved combustion completeness and reduced diesel fuel consumption.

Emission Data — 380 kVA DG Set (Engine B: Cummins, Dr. Reddy's Laboratories

Table 4

Load-Wise kWh/L Improvement Summary.

Load Range	Expected kWh/L Improvement	Observed Trend
Idle / Low (0–25%)	Strong improvement	Confirmed across platforms
Medium (25–50%)	Peak improvement	8–15% for CPCB 2; 4–13% for CPCB 4
High (50–75%)	Moderate improvement	Consistent but diminishing
Rated Load (75–100%)	Neutral or marginal	Confirmed — no artificial enhancement

Table 5

Stack Emission Results — 1010 kVA DG Set (Dr. Reddy's Laboratories, Visakhapatnam) Testing by: Lata Envirotech Services | Testing Date: 06 November 2025.

Parameter	Unit	Baseline (No H_2)	6 AMPS	10 AMPS	13 AMPS
PM	mg/Nm³	68.42	63.16	52.40	36.88
NOx	ppm	901	795.5	788.5	589.5
SO_2	ppm	37	28.5	25	20.5
CO	%	0.631	0.547	0.386	0.370
CO_2	%	9.15	7.85	7.65	6.30
O_2	%	11.76	13.08	13.09	14.10
THC (NMHC)	ppm	609	569	328	311

Table 6

Percentage Emission Reduction — 1010 kVA DG Set.

Parameter	% Reduction at 6A	% Reduction at 10A	% Reduction at 13A
PM	7.7%	23.4%	46.1%
NOx	11.7%	12.5%	34.6%
SO_2	23.0%	32.4%	44.6%
CO	13.3%	38.8%	41.4%
CO_2	14.2%	16.4%	31.1%
O_2 (increase)	+11.2%	+11.3%	+19.9%
THC	6.6%	46.1%	48.9%

Table 7

Stack Emission Results — 380 kVA DG Set (Dr. Reddy's Laboratories, Visakhapatnam) Testing by: Lata Envirotech Services | Testing Date: 05 November 2025.

Parameter	Unit	Baseline (No H_2)	6 AMPS	10 AMPS	4 AMPS
PM	mg/Nm³	59.13	51.72	32.90	20.84
NOx	ppm	274	232	202	192
SO_2	ppm	22	12	10	8
CO	%	0.245	0.216	0.212	0.208
CO_2	%	5.3	4.5	4.4	4.2
O_2	%	14.32	15.27	15.34	15.62
THC	ppm	300	296	289	272

Table 8

Percentage Emission Reduction — 380 kVA DG Set.

Parameter	% Reduction at 6A	% Reduction at 10A	% Reduction at 4A
PM	12.5%	44.4%	64.8%
NOx	15.3%	26.3%	29.9%
SO_2	45.5%	54.5%	63.6%
CO	11.8%	13.5%	15.1%
CO_2	15.1%	17.0%	20.8%
O_2 (increase)	+6.6%	+7.1%	+9.1%
THC	1.3%	3.7%	9.3%

The 380 kVA Cummins engine displayed a similar emission reduction pattern, with PM showing the largest reduction (64.8% at optimal dosing) followed by SO_2 (63.6%) and NOx (29.9%).

Emission Data — 1500 kVA DG Set (Engine C: Cummins, Turbo Energy)

The 1500 kVA Cummins engine at Turbo Energy demonstrated the most dramatic PM reductions, with an 80.1% decrease at 700 ml/min HHO. NO_2 was reduced by 59.2% and SO_2 by 75.2%. CO levels were below the limit of quantification (BLQ) at all conditions including baseline, indicating inherently low CO emissions from this engine.

At the Turbo Energy 380 kVA platform, PM was reduced by 75.0% at 600 ml/min, SO_2 by 62.9%, and NO_2 by 71.9%.

Particulate Matter Data — 250 kVA DG Set (Engine D: Neetal Laboratories, Pune)

Table 9

Stack Emission Results — 1500 kVA DG Set (Turbo Energy, Paiyanoor, Chennai) Testing by: Ekdant Enviro Services Pvt. Ltd. (NABL, ISO 9001:2015) | Testing Dates: 14–18 October 2025.

Parameter	Unit	Baseline (No H_2)	300 ml/min	500 ml/min	700 ml/min
PM	mg/Nm³	44.8	13.1	11.2	8.9
SO_2	mg/Nm³	31.4	12.9	10.3	7.8
NO_2	mg/Nm³	154.3	75.5	69.4	62.9
CO_2	%	5.6	5.1	4.8	3.2
CO	—	BLQ (< 0.2%)	BLQ	BLQ	BLQ
O_2	%	12.0	13.8	14.6	15.8

Table 10

Percentage Emission Reduction — 1500 kVA DG Set.

Parameter	% Reduction at 300 ml/min	% Reduction at 500 ml/min	% Reduction at 700 ml/min
PM	70.8%	75.0%	80.1%
SO_2	58.9%	67.2%	75.2%
NO_2	51.1%	55.0%	59.2%
CO_2	8.9%	14.3%	42.9%
O_2 (increase)	+15.0%	+21.7%	+31.7%

Table 11

Stack Emission Results — 380 kVA DG Set (Turbo Energy, Paiyanoor, Chennai) Testing by: Ekdant Enviro Services Pvt. Ltd. | Testing Dates: 14–18 October 2025.

Parameter	Unit	Baseline (No H_2)	300 ml/min	600 ml/min
PM	mg/Nm³	39.2	11.7	9.8
SO_2	mg/Nm³	23.2	13.0	8.6
NO_2	mg/Nm³	69.3	34.7	19.5
CO_2	%	4.6	4.2	3.3
CO	—	BLQ	BLQ	BLQ
O_2	%	12.8	13.4	14.3

Table 12

Stack Emission Results — 250 kVA DG Set (Saarthi GreenTech, Pune) Testing by: Neetal Laboratories and Environmental Services Pvt. Ltd. (NABL, ilac-MRA) | Testing Date: 30 August 2025 | Method: IS 11255 (Part 1).

Report No.	Load	HHO Flow	PM (mg/Nm³)	PM (Kg/kw.hr)	Flue Gas Temp (°C)	Gas Volume (Nm³/Hr)
NLES/25-26/08/ST/RE/372	80% (Without System)	None	413.0	0.403	432	977.4
NLES/25-26/08/ST/RE/373	80% (With HHO)	750 ml/min	89.9	0.083	449	926.7
NLES/25-26/08/ST/RE/377	80% (With HHO)	300 ml/min	90.9	0.088	443	969.8
NLES/25-26/08/ST/RE/374	60% (Without System)	None	1.3	0.0012	357	954.1
NLES/25-26/08/ST/RE/375	60% (With HHO)	500 ml/min	N.D.*	N.D.	365	902.2
NLES/25-26/08/ST/RE/376	60% (With HHO)	300 ml/min	N.D.*	N.D.	378	961.7
NLES/25-26/08/ST/RE/378	60% (With HHO)	160 ml/min	N.D.*	N.D.	348	1003.8

N.D. = Not Detected (below detection limit of the instrument)

The 250 kVA genset data reveals striking PM reductions. At 80% load, PM concentration decreased from 413.0 mg/Nm³ (baseline) to 89.9 mg/Nm³ with 750 ml/min HHO, representing a 78.2% reduction. With 300 ml/min HHO, PM decreased to 90.9 mg/Nm³ (78.0% reduction), indicating that the PM reduction benefit plateaus at relatively modest HHO flow rates for this engine. At 60% load, baseline PM was already very low (1.3 mg/Nm³), and hydrogen fuel assist reduced PM to below detection limits at all tested flow rates (160, 300, and 500 ml/min).

Emission Data — 500 kVA KOEL DG Set (Engine E: Cilicant Pvt. Ltd., Pune — 2023 Model Year)

Critical Observation — Evidence of Narrow Optimal Window in Newer Engines

The 500 kVA KOEL genset (2023 model year) provides the most significant evidence for the relationship between engine age/tuning state and hydrogen dosing tolerance. This engine, being newly manufactured and well-tuned, exhibited a fundamentally different dosing response compared to the older engines (2006–2018) tested in this study:

At 12–13A electrolyser current (the highest tested dosing level), exhaust O_2 *decreased* from 11.7% to 10.2%, and CO_2 *increased* from 9.16% to 9.27%. This reversal of the expected trend — where O_2 should increase and CO_2 should decrease

Table 13

A. Stack Emission Results — 500 kVA KOEL DG Set (Cilicant, Pune) Testing by: Neetal Laboratories (NABL, ilac-MRA) | Report No. NLES/25-26/12/ST/RE/1367 | Testing Date: 23 December 2025 | Sampling by: Tulsi Environmental Services & Consultant | Stack Dimension: 0.1524 m, Stack Height: 20.0 m, Stack Type: Round MS.

Parameter	Unit	Baseline (No HHO)	12–13 AMPS	9–10 AMPS	7–8 AMPS
PM	mg/Nm³	25.7	7.36	6.97	6.1
SO_2	mg/Nm³	12.4	5.46	6.20	7.36
NOx	mg/Nm³	810	687	653	682
CO_2	%	9.16	9.27	7.95	6.3
CO	%	0.587	0.57	0.41	0.35
HC	ppm	409	397	277	270
O_2	%	11.7	10.2	13.97	14.3
Flue Gas Temp	°K	368	—	—	—

Table 13

B. Percentage Change from Baseline — 500 kVA KOEL DG Set.

Parameter	Change at 12–13A	Change at 9–10A	Change at 7–8A
PM	−71.4%	−72.9%	−76.3%
NOx	−15.2%	−19.4%	−15.8%
CO_2	**+1.2% (INCREASE)**	−13.2%	−31.2%
CO	−2.9%	−30.2%	−40.4%
HC	−2.9%	−32.3%	−34.0%
O_2	**−12.8% (DECREASE)**	+19.4%	+22.2%

under effective hydrogen assist — constitutes direct experimental evidence of Zone C (over-dosing) operation. The engine consumed *more* diesel fuel with hydrogen assist at this dosing level, confirming that the hydrogen input exceeded the optimal window and transitioned into the efficiency degradation regime.

At 9–10A, the trends reversed to positive territory: O_2 increased to 13.97% and CO_2 decreased to 7.95%, indicating a transition back into Zone B operation with modest efficiency improvement.

At 7–8A (the lowest tested dosing level), the best results were observed: O_2 increased to 14.3% (+22.2%) and CO_2 decreased to 6.3% (−31.2%), with CO reduced by 40.4% and HC by 34.0%. This confirms that the optimal hydrogen dosing window for this newer, well-tuned engine lies at substantially lower hydrogen input levels compared to the older engines.

Comparison with Older Engine Platforms: In contrast, the older Caterpillar C32 TA (2006) tolerated electrolyser currents up to 13A with consistent positive results across all dosing levels (O_2 increasing and CO_2 decreasing progressively). The 2012 Cummins 1500 kVA similarly showed positive trends up to 700 ml/min HHO flow. The fact that the 2023 KOEL engine crossed into Zone C at 12–13A — a dosing level that produced the *best* results on the 2006 Caterpillar — demonstrates that the optimal hydrogen window width is inversely related to engine tuning quality and mechanical condition.

This finding has profound implications for CHFA system design: newer engines with modern fuel injection systems, precise calibration, and minimal wear require significantly lower hydrogen dosing levels and tighter adaptive control to remain within the beneficial Zone B. The margin between optimal enhancement and over-dosing degradation narrows considerably as engine condition improves.

Discussion of Emission Trends

The emission data across all four platforms reveals several consistent and scientifically significant trends:

Carbon Monoxide (CO) Reduction: CO is a direct indicator of incomplete combustion. The consistent reduction in CO across all platforms (up to 41.4% at the 1010 kVA genset) confirms that hydrogen fuel assist improves combustion completeness. Hydrogen's high flame speed and ability to ignite lean mixtures facilitates the oxidation of CO to CO_2 in the late combustion phase, reducing the escape of partially oxidised carbon species into the exhaust.

Nitrogen Oxides (NOx) Reduction: The observed NOx reductions (up to 34.6% at 1010 kVA, 29.9% at 380 kVA, 59.2% NO_2 reduction at 1500 kVA) appear counterintuitive given hydrogen's higher flame temperature. However, several mechanisms explain this finding. First, improved combustion completeness means less diesel fuel is burned to produce the same power output, reducing total heat release and peak bulk cylinder temperatures. Second, the reduced combustion duration under CHFA conditions decreases the residence time of combustion products at high temperatures, limiting thermal NOx formation via the Zeldovich mechanism. Third, the increased exhaust O_2 concentration suggests leaner overall combustion, which further suppresses thermal NOx formation.

Particulate Matter (PM) Reduction: PM showed the most dramatic reductions across all platforms, reaching 80.1% at the 1500 kVA genset and 78.2% at the 250 kVA genset. Hydrogen enhances the oxidation of soot precursors during the early combustion phase, when OH radical concentrations are elevated due to hydrogen combustion. The pre-mixed hydrogen-air charge creates a more oxidising environment that suppresses soot nucleation and promotes the oxidation of existing soot particles.

Total Hydrocarbon (THC) Reduction: THC reductions of up to 48.9% at the 1010 kVA platform confirm improved fuel utilisation. Unburned hydrocarbons in diesel exhaust primarily originate from fuel that escapes combustion in crevice volumes, lean flame-quench zones, and over-lean regions. Hydrogen's wide flammability limits enable combustion in these regions where diesel alone would not sustain a flame.

CO_2 Reduction and O_2 Increase: The simultaneous decrease in exhaust CO_2 and increase in O_2 across all platforms provides the most direct evidence of reduced diesel fuel consumption. At constant load, lower CO_2 indicates less carbon being oxidised (i.e., less diesel being burned), while higher O_2 indicates that a greater proportion of the intake air passes through the engine without participating in combustion. These trends are fully consistent with the observed kWh/L improvements.

Fuel Consumption and Energy Efficiency Metrics

Fuel consumption improvements resulting from controlled hydrogen fuel assist were quantified using BSFC and kWh per litre metrics across injection pressure regimes. Table 13 summarises representative results.

The results confirm that hydrogen assist remains beneficial across engine generations, provided dosing is adapted to fuel injection characteristics. The higher baseline efficiency of CPCB 4 engines limits the absolute magnitude of improvement but does not eliminate the benefit of controlled hydrogen addition.

Summary of Updated Findings

The experimental results across five diesel generator platforms spanning 250 to 1500 kVA, including engines manufactured between 2006 and 2023, establish the following key findings:

Table 14

Effect of Hydrogen Fuel Assist on Fuel Consumption Across Injection Pressures.

Engine Type	Load (%)	Baseline BSFC (g/kWh)	CHFA BSFC (g/kWh)	BSFC Improvement (%)	kWh/L Improvement (%)
Low Injection Pressure (CPCB 2)	25	235	210	10.6	11.2
Low Injection Pressure (CPCB 2)	50	220	198	10.0	10.5
Low Injection Pressure (CPCB 2)	75	215	205	4.7	5.0
High Injection Pressure (CPCB 4)	25	225	212	5.8	6.0
High Injection Pressure (CPCB 4)	50	210	202	3.8	4.0
High Injection Pressure (CPCB 4)	75	205	203	1.0	1.2

Table 15

Consolidated kWh/L Improvement by Engine Generation.

Engine Type	Load Range	Baseline kWh/L	CHFA kWh/L	% Change
Low Pressure — CPCB 1 or 2	40–60%	2.88–3.5	3.88	8–15%
High Pressure — CPCB 4	40–60%	3.4–3.7	3.85	4–13%

The optimal hydrogen dosing level is engine-specific and injection-pressure dependent, with lower injection pressure engines tolerating higher hydrogen fractions and exhibiting broader optimal windows. Higher injection pressure engines require tighter hydrogen regulation to maintain operation within Zone B. Atomisation quality is confirmed as a primary determinant governing hydrogen's effectiveness as a combustion enhancer. Comprehensive emission data from NABL-accredited laboratories demonstrates consistent reductions in PM (up to 80.1%), NOx (up to 34.6%), CO (up to 41.4%), THC (up to 48.9%), SO_2 (up to 75.2%), and CO_2 (up to 42.9%) under optimal hydrogen dosing conditions. The simultaneous increase in exhaust O_2 concentration across all platforms (at optimal dosing) confirms reduced diesel consumption. Critically, the 2023 KOEL 500 kVA engine demonstrated that newer, well-tuned engines have a substantially narrower optimal hydrogen window, with over-dosing (Zone C) occurring at electrolyser currents that produced the best results on older engines. This validates the hypothesis that engine age, tuning state, and injection system sophistication are primary determinants of hydrogen dosing tolerance, and that adaptive control is essential for universal applicability across diverse engine architectures, conditions, and age profiles.

End of Part 1 - The remainder of the paper (from Section 5 onwards) appears in Part 2. The continuation of results and advanced combustion analysis is presented in Part II of this series.

Declarations

1. Conflict of Interest: All authors declare that they have no conflicts of interest.
2. Funding: No funds, grants, or other support were obtained.
3. Author Contributions: All authors contributed equally.

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